Week 1 Notes

STAT 251 SECTION 01

Introduction to statistical methods

Where does statistics come from?

Statistics began as bookkeeping about people. Long before it was a mathematical discipline, it was the practice of counting births, deaths, and taxpayers so that a state could describe itself - which is also where the word comes from.

- **1662 - John Graunt.** Working from London's weekly *Bills of Mortality*, Graunt classified deaths by cause, compared urban and rural mortality, and built the first life table. It is generally regarded as the first statistical analysis of data.
- **1693 - Edmond Halley.** Using birth and death records from the city of Breslau, Halley produced a life table complete enough to price life annuities, giving governments (and later insurers) a way to put a number on risk.
- **1749 - Gottfried Achenwall.** Achenwall coined the German word *Statistik* for the systematic description of a state. Sir John Sinclair carried "statistics" into English in his *Statistical Account of Scotland* (1791).
- **1790 - The first national censuses.** The United States conducted its first census in 1790; Great Britain followed in 1801. Counting the population became a routine function of government.
- **1830s - Adolphe Quetelet.** Quetelet applied the normal curve to human characteristics such as height and chest circumference, introducing *l'homme moyen* - "the average man" - and moving statistics from state record-keeping into the social sciences.
- **Around 1900 - Francis Galton and Karl Pearson.** Galton and Pearson developed correlation and regression, and Pearson introduced the chi-square test, turning statistics into a mathematical discipline in its own right.
- **1908-1935 - W. S. Gosset and R. A. Fisher.** Gosset (writing as "Student") derived the t-distribution while working at the Guinness brewery. Fisher then formalized experimental design, analysis of variance, and significance testing in *Statistical Methods for Research Workers* (1925) and *The Design of Experiments* (1935). Most of what we cover this semester comes from this period.

- **Today.** The ideas are largely the same; computers let us apply them to data sets far larger than anything Graunt could count by hand.

Sources: J. Graunt (1662), *Natural and Political Observations Made upon the Bills of Mortality*. E. Halley (1693), *Phil. Trans. R. Soc.* 17:596-610. G. Achenwall (1749), *Abriss der neuesten Staatswissenschaft*. A. Quetelet (1835), *Sur l'homme et le developpement de ses facultes*. R. A. Fisher (1925), *Statistical Methods for Research Workers*.

Data, observations, variables

- **Statistics** is a scientific field that deals with the collection, description, analysis, interpretation, and presentation of data. We see and use data in nearly all aspects of our everyday lives: politics, medicine, forecasting, finance, marketing, etc. In short, we use statistics to ask and answer scientific questions and make predictions. To put it another way, statistics is the science of **data**
- **data** are pieces of factual information, such as measurements which are recorded and used for the purpose of analysis. In summary, data is the raw information from which statistics are created. Statistics are the results of data analysis - their interpretation and presentation.
- Data can be either **structured** or **unstructured**, depending on whether or not they arrive in a predefined format.
  - **structured data** - data that follow a predefined format with fixed rows, columns, and fields, so that every observation is recorded in the same way. Structured data can be stored directly in a spreadsheet or database and analyzed without any additional processing. Examples include spreadsheets and CSV files, survey responses, sensor readings, and transaction records.
  - **unstructured data** - data that have no predefined format or fixed fields. Unstructured data must be processed and organized into a structured form before they can be analyzed. Examples include free-text notes, emails, and social media posts; images, audio, and video; and web pages and PDF documents.
  - Nearly all of the methods in this course assume the data are structured - that is, that they have already been arranged into the kind of data table described below.
- Typically data are represented in a square table called a data table. The fundamental unit of data is an **observation** - a single **row** of a data table representing a set of measurements on one or more **variables**.
- A **variable** is a characteristic of an observation. In a typical data table, the variables are represented by the **columns** of the data table.

The following example is 20 of the 400 stores in the `Carseats` data set, which records sales of child car seats. For each store the unit sales, price charged, community income level, regional population, local education level, shelving location quality, and whether the store is in an urban location and in the U.S. are recorded.

Observation	Sales (1,000s)	Price ($)	Income ($1,000s)	Population (1,000s)	Education (years)	ShelveLoc	Urban	US
1	4.38	108	98	173	16	Bad	Yes	No
2	11.62	139	83	325	17	Good	Yes	Yes
3	5.07	96	91	334	17	Bad	Yes	Yes
4	6.87	131	105	249	13	Medium	Yes	Yes
5	11.27	133	68	60	16	Good	Yes	Yes
6	3.67	131	31	327	16	Medium	Yes	No
7	4.19	112	93	420	11	Bad	Yes	Yes
8	4.97	160	67	27	17	Medium	Yes	Yes
9	4.36	123	41	357	14	Bad	No	Yes
10	8.69	101	64	68	16	Medium	Yes	Yes
11	8.65	120	76	218	14	Medium	No	Yes
12	9.16	156	105	435	14	Good	Yes	Yes
13	6.37	132	77	86	18	Medium	Yes	Yes
14	4.21	137	35	502	10	Medium	No	Yes
15	8.57	99	78	158	11	Medium	Yes	Yes
16	6.80	135	117	337	10	Bad	Yes	Yes
17	8.61	102	38	283	15	Medium	Yes	No
18	10.26	108	75	377	12	Good	Yes	No
19	7.90	124	46	206	11	Medium	Yes	No
20	6.81	125	61	263	12	Medium	No	No

Source: `Carseats` data set, R package `ISLR2`. James, G., Witten, D., Hastie, T., and Tibshirani, R. (2021). *An Introduction to Statistical Learning*, 2nd ed. Springer. `Carseats` is a simulated data set of 400 stores distributed with the textbook; the 20 stores shown here were drawn from it.

The following are 15 observations of a study done on the morphology of Egyptian skulls from 5 epochs of Egyptian history. For each skull, the epoch, and several measurements characterizing the shape of the skull are recorded.

Epoch	maximal breadth	basibregmatic height	basialveolar length	nasal height
c4000BC	132	136	100	50
c3300BC	134	130	93	54
c200BC	136	138	94	55
c3300BC	131	136	99	56
cAD150	136	133	97	51
cAD150	140	135	103	48
c1850BC	138	133	91	46
c200BC	137	134	107	54
cAD150	143	126	88	54
c200BC	141	128	95	53
cAD150	138	125	99	51
c200BC	140	134	90	51
c1850BC	137	139	97	50
c4000BC	138	135	100	55
c4000BC	139	130	108	48

Types of variables

- Variables are distinguished by the type of information they represent. One basic distinction is between **qualitative** variables and **quantitative** variables.

- **qualitative variables** - also called categorical variables, represent non-numeric qualities or characteristics that can be placed in distinct categories. What are the qualitative variables in the `Carseats` data above?

- **quantitative variables** - Sometimes referred to as numeric variables, are numerical characteristics that represent measurements with meaningful magnitudes (their values can be ordered and the differences between them are meaningful). In the Egyptian skull data which variables are quantitative variables?

- Qualitative variables can be further divided between **nominal** and **ordinal** variables.
  - **nominal variables** - non-numeric qualities or characteristics that can be placed in distinct categories that do not have a natural ordering. Examples of variables that cannot be ordered are gender, race, eye color, or political party. In the `Carseats` data, which variables would be considered nominal variables?
  - **ordinal variables** - non numeric qualities or characteristics that can be placed in distinct categories with an inherent ordering. Examples include education level (e.g., bachelors, masters, PhD) or temperatures (e.g., cool, warm, hot).
    - **Likert scale** responses are a common example of ordinal variables that are typically found in surveys. A **Likert scale** is a rating scale used to measure opinions, attitudes, or behaviors. It consists of a statement or a question, followed by a series of five or seven answer statements. Respondents choose the option that best corresponds with how they feel about the statement or question.
- Quantitative variables can also be further divided between two sub-categories: **discrete** and **continuous** variables.
  - **discrete variables** - are quantitative variables that take on distinct, countable values (e.g., whole numbers or integers) such $0, 1, 2, 3. \ldots$. Any quantitative variable that represents counts of objects/items are quantitative discrete. For example, age is a count of the number of years someone/something has been alive.
  - **continuous variables** - are variables that can take on infinite number of values within an interval of any two specific values (e.g., temperature $^{\circ}C/^{\circ}F$, height in inches, speed in miles per hour).

Populations and samples

- Statistics is generally concerned with studying properties of a **population**. You can think of a population as a collection of all possible persons, things, or objects that you are studying. Another way to think of a population is as a collection of all possible observations of a variable - both observed and unobserved.
    - A population can be finite (countable) or infinite (uncountable). Many populations are so large that we treat them as if they are infinite. For example, the population of students taking STAT 251 Section 01 is a finite set of 60 individuals. Alternatively, the number of germs living in a patient's body is effectively infinite.
- For many populations, it is very difficult or even impossible to observe all members of the population. To tackle this challenge statistics uses **samples** to learn about the properties of a population. A **sample** is a subset of the population that is actually observed. You can think of a sample as a set of the observed observations.
    - The idea of sampling is to select a portion of individuals or objects that are representative of the population
    - To illustrate the distinction between a sample and a population, let's take the example of a wildlife biologist researching the paw size of mountain lions in the state of Idaho. In this scenario, the population being examined encompasses all the mountain lions residing in the state, totaling approximately 2,000 individuals. However, due to the considerable challenges and expenses associated with tracking and observing every existing member, the biologist opts to document the paw size of a more manageable subset of 20 mountain lions. The 20 mountain lions for which the biologist records paw sizes constitute a sample from the larger population of mountain lions.
- In statistics we typically represent the size (i.e number of observations) of a population with the letter N and the size of a sample with the letter n. In the above example of the wildlife biologist studying mountain lion paw size, the population has size $N = 2,000$ and the sample the biologist took was of size $n = 20$.

A quick note on mathematical notation

Before we introduce our first formulas, it is worth taking a moment to get comfortable with some of the mathematical notation we will use throughout the course. Statistics uses a compact "shorthand" to describe calculations on data, and once it becomes familiar, formulas that look intimidating at first become easy to read. If you have never seen this notation before, don't worry - we will assume you know very little and build it up from scratch.

- **Representing observations with symbols.** Suppose we collect a sample of n observations of some quantitative variable - for example, the paw widths of $n = 20$ mountain lions. Rather than writing out every measurement, we represent the variable with a single letter (usually x or y) and attach a small subscript to refer to the individual observations:

$$x_1,\ x_2,\ x_3,\ \ldots,\ x_n$$

  Here x_1 is read as "x sub one", x_2 as "x sub two", and so on. The subscript is simply an *index* - a label that tells us <u>which</u> observation we are referring to. The symbol x_i (read "x sub i") is a stand-in for a *generic* observation, where the index i can be any whole number from 1 up to n. So whenever we write x_i, you can read it as "the i-th observation in our sample."

    - For example, if our sample of paw widths (in inches) is $\{4.2,\ 3.9,\ 5.1\}$, then $n = 3$ and $x_1 = 4.2$, $x_2 = 3.9$, and $x_3 = 5.1$.

- **The order does not matter.** It is important to understand that these subscripts are just labels - they do <u>not</u> imply any ranking, ordering, or importance. The observation we happen to call x_1 is not "first", "smallest", or special in any way; it is simply the one we chose to label with a 1. If we shuffled the observations and relabeled them, we would still have the exact same sample. In other words, $\{4.2,\ 3.9,\ 5.1\}$ and $\{3.9,\ 5.1,\ 4.2\}$ describe the same data.

- **Remembering n and N.** As we saw in the previous section, n is the number of observations in a *sample* and N is the number of observations in the *population*. These letters simply tell us how many observations we are working with. When a formula is applied to a sample it will run up to n; when it is applied to an entire population it will run up to N.

- **Summation notation.** Very often we need to add up all of the observations in our data. Writing out $x_1 + x_2 + x_3 + \cdots + x_n$ every time would be tedious, so we use a shorthand called **summation notation**, denoted by the capital Greek letter sigma, Σ (think of the "S" in "Sum"). The sum of all n observations in a sample is written

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + x_3 + \cdots + x_n$$

  which we read as "the sum of x_i, for i equals 1 to n." Let's break down each piece:

    - The expression *below* the Σ (here $i = 1$) tells us which letter is doing the counting and the **starting** value of that index.

    - The value *above* the Σ (here n) tells us the **ending** value of the index.

- The expression to the *right* of the Σ (here x_i) is the quantity we add up. We plug in $i = 1$, then $i = 2$, and so on all the way up to $i = n$, adding the results together as we go.
- For example, using our paw-width sample $\{4.2,\ 3.9,\ 5.1\}$ from above:

$$\sum_{i=1}^{3} x_i = x_1 + x_2 + x_3 = 4.2 + 3.9 + 5.1 = 13.2$$

- Summation notation is flexible, and we can add up other quantities in the same way. For instance, $\sum_{i=1}^{n} x_i^2$ means "square each observation and then add up the squares." We will encounter expressions like this later on when we measure how spread out our data are.

With this notation in hand, we are ready to write down our first statistics.

Statistics vs. Parameters

- Statistics primarily deals with **estimation** – the process of inferring an unknown quantity about a population using a set of sample data. An **estimator** is a mathematical function that **estimates** a given **parameter** based on observed data.
  - Therefore, a **statistic** is a numerical characteristic of a sample that represents a property of the population called a **parameter**. Parameters are numerical characteristics of a population that can be estimated via a statistic.
- We will use Greek letters such as θ, σ, μ to represent parameters. These symbols with "hats" like $\hat{\theta}, \hat{\sigma}, \hat{\mu}$ are used to represent estimators/statistics.
- Two statistics that we will deal with in this course are the **sample mean** and **sample proportion**.
  - A **proportion** describes the fraction of a whole that represent some property or category. It can be expressed as a value between 0 and 1 or as a percentage. The **sample proportion** $\hat{p}$ is the fraction of observations in the sample that represent a particular category. The *population proportion* is the fraction of observations in the population that represent a particular category. The sample proportion is defined mathematically as

$$\hat{p} = \frac{\text{Number of observations in category}}{n}$$

The population proportion is defined mathematically as

$$p = \frac{\text{All observations in category}}{N}$$

The sample proportion $\hat{p}$ is used to estimate the population proportion p

- The **arithmetic mean** (sometimes called average) describes the center of a population or set of data. The **sample mean** $\bar{x}$ describes the central tendency of the observations in the sample. The *population mean* describes the central tendency of the entire population. The sample mean is defined mathematically as

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n}$$

The population mean is defined mathematically as

$$\mu = \sum_{i=1}^{N} \frac{x_i}{N}$$

The sample mean $\bar{x}$ is used to estimate the population mean μ

- The tools for estimation allow us to approximate almost everything about populations using only samples. With estimates we can:
  - Assess differences among groups and relationships between variables.
  - Describe populations. Examples of estimates include averages, proportions, measures of variation, and measures of relationship.
  - Then we can ask and answer questions or formally, test and evaluate hypotheses.

Descriptive vs Inferential Statistics

Recall that statistics is a science that deals with the collection, description, analysis, interpretation, and presentation of data. The collection of data is referred to as the **design**. In statistics, design concerns the formulation of statistical questions and the process/method in which we plan to collect data to answer our statistical question.

- **Descriptive statistics** - refers to a organizational step that occurs before we do any analysis or make any decisions/predictions. It includes describing the observations in a sample using statistics or describing a population using parameters. It can also include graphical summaries of a sample or population.

- **Inferential statistics** - refers to using a sample (usually a statistic) to answer a question about a population such as estimating the value of a parameter. It includes estimating parameters, making decisions, and prediction.
- **Key Point:** Descriptive statistics can be applied to samples or populations. Inferential statistics are only applied to samples in order to make inferences about a population.